

Prohori (Probhari)

upay Scam Shield - Dataset and Scenario Specification

Track 01: Trust and Risk Intelligence | AI Dev Fest 2026 AI Hackathon (DIU CPC x upay)

What this document is	A complete, hand-over-ready specification for generating a realistic, Bangladesh-specific synthetic MFS dataset: table schemas, volumes, normal-behaviour model, fraud typologies, 18 worked scenarios with exact numbers, warning/alert copy, acceptance tests and a ready-to-paste generator prompt.
How to use it	1) Read Part 1 (assumptions). 2) Paste Parts 3-5 + Part 9 into Claude to generate the CSVs. 3) Run the acceptance tests (Part 8) before training any model. 4) Use Part 6-7 scenarios as your demo script and UI copy.
Data policy	100% synthetic. No real phone numbers, NIDs, names or upay data. Every assumption is listed so it can be copied into your README (rulebook requires dataset disclosure).

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Part 1 - Research basis, assumptions and what you must verify

Before generating data, separate **what is documented** from **what is design choice**. Judges reward teams that say clearly which is which. Items marked **VERIFY** were gathered from secondary sources and may have changed; check the primary source and write the 'as of' date in your README.

1.1 Regulatory frame that the data must respect

Rule / fact	Used in dataset as	Status
Bangladesh Bank MFS Regulations 2022: limited-purpose accounts, one personal account per person per provider, KYC-tiered limits	kyc_level (1/2/3) on every customer; per-tier daily/monthly caps enforced in normal data	Documented; exact numbers VERIFY
Agents may do cash-in/cash-out only; third-party transfer through an agent is not allowed	A P2P transfer 'at agent' is an invalid/suspicious pattern (rogue-agent signal S5)	Documented (secondary source) VERIFY
Providers must monitor transactions/agent patterns and report suspicious activity to BFIU	Justifies the analyst alert queue and case report ('what happened / why risky / what next')	Documented
upay Send Money is free; cash-out from agent carries a percentage fee; ATM cash-out is cheaper	fee_tk column: P2P 0, agent cash-out ~1.4% (config), ATM ~0.8% (config)	Reported 2021-22 era; VERIFY on upay limits-and-charges page
Per-transaction send and cash-out maximums and daily/monthly caps	Kept in config.yaml so you can change them in one place, not hard-coded	VERIFY - my quick check of public pages returned only international-remittance limits, which are NOT the same as domestic P2P limits

Suggested config defaults (placeholders - replace after verification): KYC-1 daily 10,000 / monthly 25,000 / max balance 25,000 Tk; KYC-2 25,000 / 50,000 / 50,000 Tk; KYC-3 50,000 / 200,000 / 100,000 Tk; P2P max per transaction 50,000 Tk; agent cash-out max per transaction 30,000 Tk.

1.2 Fraud evidence you can quote (VERIFY against the primary report before putting in slides)

- Policy Research Institute (PRI) Bangladesh survey of MFS users and agents (about 7,300 users and 2,000 agents, as reported in earlier research for this project): roughly 1 in 10 users reported being a fraud victim; average loss above Tk 9,000; compromised PIN and impersonation was the largest category (about 56%); agent victims lost about Tk 18,000 on average; Sylhet Division showed the highest propensity.
- Fear of being cheated was reported as a main reason non-users avoid MFS - this is your 'trust is the adoption bottleneck' story.
- Most MFS traffic in Bangladesh is on USSD / feature phones, so the warning must work as a short text and voice message, not only as an app pop-up.

1.3 Design choices (not facts)

- **Fraud is over-represented** (0.5% of rows) compared with real life, so models have enough positives. Say this openly and report PR-AUC, recall at fixed false-alarm rate - not accuracy.
- **Risk score is not a calibrated probability**. The dialog shows 'Fraud risk: 89%' only after you calibrate (isotonic or Platt scaling on the validation set). Until then label it 'Risk score 89/100'.
- Scenario numbers (e.g. '34 unique senders') are illustrative recipes chosen to be learnable and demo-able, not statistics.
- Fraud labels are assigned by the generator (ground truth). Real upay would have noisy labels - mention as a limitation.

Part 2 - Dataset overview and volumes

Item	Target	Notes
Simulation window	60 days (e.g. 1 Aug - 29 Sep 2026)	Includes salary window (1st-5th), Fridays, one Eid-like surge week (set a fictional date)
Customers	10,000 (incl. ~120 fraud-controlled wallets)	Fraud wallets are normal-looking rows in customers.csv; role only in labels
Agents	300 in 30 peer groups (division x area type)	3-5 rogue agents injected
Transactions	about 600,000 rows	Roughly 10,000 per day; USSD ~50%, app ~30%, agent-assisted ~20%
Fraud rows	about 3,000 (0.5%)	Row-level label; see class mix below
Train / valid / test	Days 1-40 / 41-50 / 51-60	Time-based split; test locked; scenarios of every class in each split
Random seed	42	Same seed reproduces identical files

2.1 Fraud class mix (row level, approximate)

Class	Name	Share	Approx. rows	Approx. cases	Rows per case
S1	OTP / PIN takeover (fake customer care)	45%	1,350	~900 victims	1-3
S2	Prize / lottery / fake-job fee to collector wallet	20%	600	~12 collectors x ~35 victims	1 per victim + cash-outs
S3	Money-mule chain / collector funnel	15%	450	~25 chains	4-30
S4	Fake seller (advance payment)	12%	360	~15 sellers	~20 buyers each + cash-out
S5	Rogue agent	5%	150	4 agents	Excess cash-outs
S6	SIM-swap takeover	3%	90	~45 victims	1-3
S7	Guided-victim (elderly / first-time user coached by caller)	included in S1/S2 rows	-	-	Behavioural variant

Part 3 - Table schemas

Keep **raw tables**, the **label table** and **derived features** separate. Models must never see the label table; derived features are computed by `src/features.py` from raw data only, using only information available at the time of the transaction (no future leakage).

3.1 customers.csv (10,000 rows)

Column	Type	Allowed values / rule	Example
customer_id	str	C000001...C010000	C004211
phone_hash	str	SHA-256 of fake number; never a real number	a91f...c3
age	int	16-75; skew 20-45	34
gender	str	M 58% / F 42%	M
occupation	str	rickshaw/van driver, garment worker, small trader, farmer, student, housewife, migrant-worker family, service holder, day labourer, shop owner	rickshaw driver
division	str	8 divisions; Dhaka ~30%, Chattogram ~20%, others share the rest; Sylhet slightly higher fraud exposure	Dhaka
area_type	str	urban 40% / semi-urban 25% / rural 35%	urban
kyc_level	int	1 = 40%, 2 = 35%, 3 = 25% (config)	2
registration_date	date	Before window start for 95%; 5% register during window (new-wallet legit cases)	2024-03-18
device_type	str	smartphone 30% / feature phone 20% / USSD-only 50%	feature phone
digital_literacy	str	low 40% / medium 40% / high 20% (used for victim-selection probability)	low
income_band	str	low / lower-mid / mid; drives median transaction size	low
home_agent_id	str	Nearest agent id (cash-in/out habit)	A0123

3.2 agents.csv (300 rows)

Column	Type	Rule	Example
agent_id	str	A0001...A0300	A0123
division / area_type	str	Same lists as customers	Dhaka / urban
peer_group_id	str	division x area_type bucket (about 10 agents each)	PG-DHK-U-03
daily_cashout_capacity_tk	int	Urban 400k-900k, semi 200k-500k, rural 100k-300k	520000
typical_daily_cashouts	int	Peer-group mean +/- 25% (normal agents)	46
registration_date	date	Before window	2023-07-02

3.3 transactions.csv (about 600,000 rows)

Column	Type	Rule	Example
txn_id	str	T00000001...	T00412907
timestamp	datetime	Asia/Dhaka, second resolution, sorted ascending	2026-09-12 21:14:07
sender_id / recipient_id	str	customer_id, agent_id (for cash-in/out) or merchant id M0001...	C004211 / C009307
txn_type	str	P2P_SEND, CASH_IN, CASH_OUT_AGENT, CASH_OUT_ATM, MERCHANT_PAY, BILL_PAY, AIRTIME, ADD_MONEY, REMITTANCE_IN, G2P_ALLOWANCE	P2P_SEND
amount_tk	float	> 0; respects KYC per-txn and daily caps (except labelled violations)	20000
fee_tk	float	P2P 0; agent cash-out ~1.4%; ATM ~0.8%; others 0	0
sender_balance_before/after	float	Consistent: after = before - amount - fee	23400 / 3400
recipient_balance_before/after	float	Consistent: after = before + amount (null for agents/merchants)	310 / 20310
channel	str	APP / USSD / AGENT_ASSISTED	APP
agent_id	str	Required for cash-in/out; null otherwise	null

Column	Type	Rule	Example
device_id	str	Stable per customer, e.g. D-C004211-1	D-C004211-1
device_changed	0/1	First time this device_id appears for the sender	0
location_changed	0/1	District differs from sender's usual district	0
session_seconds	int	Time from login/USSD start to confirm; guided victims are long (150-600 s)	42
pin_attempts	int	1 normally; 2-4 for coached or confused users	1
counterparty_first_time	0/1	Sender never transacted with recipient before	1

3.4 labels.csv (ground truth - NEVER given to the model as a feature)

Column	Type	Rule	Example
txn_id	str	Join key	T00412907
is_fraud	0/1	Row is part of a fraud flow	1
fraud_class	str	S1-S7 or null	S3
fraud_role	str	victim_send, takeover_send, collector_inbound, mule_hop, cashout, rogue_agent_cashout	victim_send
case_id	str	Groups rows of one scam (e.g. one collector and all its victims)	CASE-S3-014
scenario_id	str	SC-01...SC-10 or SB-01...SB-08 for hand-planted showcase cases	SC-01
is_benign_lookalike	0/1	Legit rows deliberately resembling fraud (Part 7)	0
expected_band	str	ALLOW / NUDGE / WARN / HIGH - used by acceptance tests	HIGH

3.5 Derived features (computed by the pipeline, not stored in raw CSVs)

Feature	Definition (as of txn time)	Fraud signal
amount_to_median_ratio	amount / sender's 30-day median send amount	Sudden jump (SC-01: 10x)
amount_to_kyc_cap	amount / sender's per-txn or daily cap	Near-cap amounts
balance_drain_ratio	amount / sender_balance_before	Account emptying (S1, S6)
hour_deviation	Distance from sender's usual active hours	3 AM activity (S1)
is_new_device / location_changed	From raw columns	Takeover
counterparty_first_time	From raw	Stranger payment
recipient_age_hours	Time since recipient's first appearance or registration	Fresh collector wallet
recipient_unique_senders_3h / 24h	Distinct senders to recipient in window	Funnel (S2-S4)
recipient_first_time_sender_ratio	Share of inbound senders who are first-time	Victim pool
recipient_inflow_outflow_lag_min	Minutes from inbound to outbound/cash-out	Pass-through (S3)
recipient_cashout_ratio	Cash-out amount / inbound amount in 24h	Cash-out urgency
recipient_supplier_outflow_count	Outflows to merchants / wholesalers	Zero for fake sellers (S4)
pin_attempts, session_seconds	From raw	Coached victim (S7)
agent_peer_zscore	(agent daily cash-outs - peer mean) / peer std	Rogue agent (S5)
agent_flagged_wallet_share	Share of agent cash-outs from wallets later flagged	Collusion

3.6 Behaviour profile per customer (generated first, used to draw normal transactions)

Profile field	How to draw it	Example
median_send_tk	Lognormal by income_band: low ~ Tk 700-1,500, lower-mid ~ 1,500-3,000, mid ~ 3,000-6,000	2,000 (SC-01 sender)
sends_per_week	Poisson mean 1.5 (low) to 5 (shop owner)	2
regular_contacts	3-8 recipients; 80% of P2P sends go to them	6
active_hours	Two peaks (e.g. 12-14 and 19-22); night (00-05) probability under 2%	19-22
usual_channel / agent	From device_type; home_agent_id for cash in/out	USSD

Part 4 - Normal behaviour model

Fraud detection only works if normal life looks like normal life. Use these rules for the 99.5% of rows that are legitimate.

Behaviour	Rule	Why
Transaction mix	P2P_SEND 38%, CASH_OUT_AGENT 18%, CASH_IN 17%, AIRTIME 10%, BILL_PAY 7%, MERCHANT_PAY 5%, others 5%	P2P and cash dominate; savings and merchant use are minorities
Amount distribution	Heavy-tailed lognormal around each customer's median; 90% of sends under 5,000 Tk; sends above 15,000 Tk are under 1% of normal P2P	Large sends are naturally rare, which makes them informative
Recipient regularity	80% of P2P to a small contact list; 20% to others (new contact rate ~5% per month)	First-time counterparty is a mild signal, not proof
Weekly rhythm	Friday and weekend uplift in family transfers (+35%); Thursday evening cash-outs	Remittance and wage habits
Salary window	Days 1-5: cash-in volume x1.8 for garment and service workers; cash-out in the following days	Wages paid at month start
Bill window	Days 20-28: BILL_PAY x2	Utility due dates
Eid-like surge	One 5-day window: volume x2.5, P2P amounts x2-3 for 40% of customers (gifts, family)	Legit large sends must exist or the model learns 'big = fraud'
Time of day	Peaks 08-10, 12-14, 19-22; 00-05 under 2%	Night is rare
Balances	Median balance 1,200 Tk (low), most wallets under 10,000 Tk; refill by cash-in before a large send	Balance consistency checks
New legit wallets	5% of customers register during the window; behave normally; 30% receive from 3-6 contacts in the first week	Prevents 'new = fraud' shortcut
Merchants / shop owners	~300 small-trader wallets receive 20-60 payments/day from regular customers; wallet age > 6 months	Legit funnel shape (SB-04)
Remittance / G2P	REMITTANCE_IN and G2P_ALLOWANCE are one-to-many disbursements from a few trusted source ids	Legit fan-out (SB-08)

Part 5 - Fraud typologies and injection recipes

Each class has a **recipe** the generator must follow. Randomise within the ranges so models learn the pattern rather than memorise one number.

Class	Story	Injection recipe (ranges)	Strongest signals
S1 OTP/PIN takeover	Caller poses as upay/bank staff, gets PIN/OTP, drains the account	Victim: low/medium literacy. Event hour 22:00-04:00 for 60%, else any. device_changed=1, location_changed 50%. Amount = 70-98% of balance (balance 3k-30k). Recipient wallet age 2-72 h. 1-3 sends within 15 min, sometimes followed by recipient cash-out within 30 min.	drain_ratio, new device, odd hour, young recipient
S2 Prize/lottery/fake-job fee	'You won' or 'job registration fee'; victim pays a small amount	12 collector wallets (age 0.5-3 days). Each gets 15-40 victim sends of 200-3,000 Tk over 6-48 h; 85% first-time senders. Collector cash-outs 80-95% of inflow within 1-3 h.	Collector features; individual amount looks normal
S3 Mule chain / collector funnel	Organised group moves stolen funds through several wallets	~25 chains: source -> 2-4 mule wallets -> agent cash-out. Hop delay 2-15 min, forward 90-99% of amount (small skim). Wallets age 1-5 days. Some chains split into 2-3 branches.	Pass-through lag, graph shape
S4 Fake seller	Facebook/WhatsApp shop takes advance payments and vanishes	~15 seller wallets (age 3-20 days). 8-30 buyers each, 500-6,000 Tk, 90% first-time. Zero outflows to suppliers/merchants; cash-out via 1-2 agents; account silent afterwards.	In-degree, zero supplier outflow
S5 Rogue agent	Agent processes cash-outs for stolen wallets or over-charges	4 agents with 7-10x peer cash-out count for 5-10 days; 40-60% of excess from wallets flagged in S1-S4; many round amounts just under cap; some invalid third-party transfers at agent.	agent_peer_zscore, flagged-wallet share
S6 SIM swap	Fraudster duplicates SIM, takes over wallet	~45 victims: long quiet period then at a new device_id, 02:00-06:00, 60-95% balance drain; no pin_attempts failure; recipient age under 48 h. Keep detection hard (borderline scores 55-75).	Device + drain; low separability by design
S7 Guided victim	Elderly/first-time user coached on the phone by a scammer	USSD or feature phone, session 150-600 s, pin_attempts 2-4, amount 5-15x median, recipient young, often 'cash-out' at agent right after.	session_seconds, pin_attempts

Rules that keep the data honest

- Every fraud row gets is_fraud=1, a class, a role and a case_id; victims' normal rows stay labelled 0.
- About 20% of fraud cases should be **weak or ambiguous** (e.g. small amount, daytime, known device) so the model cannot reach 100%.
- Add 'near-miss' legit rows (Part 7) at about 1.5x the number of fraud cases so false-alarm rate is realistic.
- Respect balances: a victim cannot send more than they have; use cash-in first if you need a large balance.
- Cash-out amounts and fees must follow the config (agent max 30,000 Tk per transaction -> split big cash-outs into several).

Part 6 - Scenario library (fraud)

Each scenario is a hand-planted showcase case: put it in the dataset with a known `scenario_id`, use it for the live demo, and use it as an acceptance test. Fusion formula: **score = 100 x (0.5 x P(fraud) + 0.3 x anomaly + 0.2 x graph)**, rounded half up. Component values shown are the *expected* model outputs, not guarantees.

SC-01 - Normal user sends 10x to a fresh collector wallet (your example)

Behaviour: a stable low-value user suddenly tries a large send to a stranger's wallet that was opened days ago, has collected from many strangers and cashes out almost instantly. This is the classic S2/S3 collector pattern seen from the victim's side - and the exact moment Prohori must interrupt.

Fact	Value
Sender	Rahim (C004211), 34, rickshaw driver, KYC-2, feature phone + app. Median send 30 days: 2,000 Tk ; typical 1-2 sends/week to family and a mobile-recharge shop; 98% of sends under 4,000 Tk.
Event	Fri 21:14: tries P2P_SEND 20,000 Tk (10x median) to W-77310. Never sent to it before. Balance before: 23,400 Tk.
Recipient W-77310	Registered 3 days (68 h) ago . Inbound in last 48 h: 31 transfers from 29 unique senders (24 first-time), total 61,500 Tk. Outflow: 3 cash-outs (29,000 + 29,000 + 2,400 Tk) at two agents , each within 3-12 minutes of an inbound burst.
Context	Sender's last big send ever: 4,500 Tk (Eid). Session 38 s; PIN entered first time.

Model signals: amount_to_median_ratio 10.0; counterparty_first_time=1; recipient_age_hours 68; recipient_unique_senders_48h 29; first-time sender ratio 0.83; recipient_inflow_outflow_lag 3-12 min; recipient_cashout_ratio 0.96. SHAP top reasons: recipient age, unique senders, amount vs median.

Fusion: $0.5 \times 0.90 + 0.3 \times 0.80 + 0.2 \times 1.00 = 0.89 \rightarrow$ **89/100, band HIGH / HOLD**. Highest friction: warning, extra confirmation and a short cooling-off.



Customer-facing alert

Stop and check! Fraud risk: HIGH (89/100).

This number was opened only **3 days ago**. In the last 2 days **29 different people** sent money to it, and most of it was **cashed out within minutes**. You are sending **20,000 Tk** - 10 times more than you usually send. People who do this often lose their money to scams. **Do you personally know and trust this person?**

[Cancel (recommended)] [I know them - continue after 10-minute check]

upay never asks for your PIN or OTP by phone.

Bangla (transliterated): Sabdhan! Ei number-ti matro 3 din age khola hoyeche. Gato 2 dine 29 jon bhinno manush ekhane taka pathiyeche ebong prai sathe sathe cash-out hoyeche. Apni ager cheye 10 gun beshi pathacchen. Apni ki ei byaktike byaktigotobhabe chenen?

Analyst case report: Wallet W-77310 (age 68 h) received 61,500 Tk from 29 unique senders in 48 h (83% first-time) and cashed out 60,400 Tk at agents A0457 and A0458 within 3-12 min of each inbound burst. Pattern matches a collector/mule wallet. Customer C004211 attempted 20,000 Tk (10x median); customer was warned and cancelled. **Recommended:** freeze W-77310, review agents A0457/A0458, contact the 29 senders, escalate to BFIU reporting workflow.

SC-02 - OTP/PIN takeover at night (S1 - the biggest real-life class)

Behaviour: the owner never acts like this - wrong hour, new device, new district, almost the whole balance. The account owner may not even know it is happening, so the control must be **step-up verification** (call-back / secret question / SMS to the registered contact) plus a warning to the legitimate owner.

Fact	Value
Victim	Salma (C002907), 27, garment worker, KYC-1, smartphone, median send 800 Tk, balance 14,200 Tk. Always active 19:00-22:00 from the same phone.
Event	02:47 a login from a new device in a different district. Tries 12,500 Tk (88% of balance) to W-90022, wallet age 11 h. 2 minutes earlier she had a 4-minute phone call (not in data, implied by session_seconds and OTP request count).
Planted rows	3 rows: OTP request, send 12,500, second send of 1,500 within 4 min (S1 takeover_send).

Model signals: is_new_device=1; location_changed=1; hour_deviation 5.1 sd; balance_drain_ratio 0.88; recipient_age 11 h; amount_to_kyc_cap 1.0 (KYC-1 daily cap use).

Fusion: $0.5 \times 0.90 + 0.3 \times 0.85 + 0.2 \times 0.50 = 0.81 \rightarrow$ **81/100, band WARN.** WARN band - block-until-verified because device is new.

▼ Risk score 81/100



Customer-facing alert

Security check - is this you? Risk 81/100.

A new phone is trying to send **12,500 Tk** at **2:47 AM**. If this is not you, do not share any OTP. We will call your registered number to confirm.

[This is not me - lock my wallet] [Yes, it is me]

Bangla (transliterated): Raat 2:47-e notun phone theke 12,500 taka pathanor chesta hocche. Eti ki apni? Na hole OTP kauke dien na.

Analyst case report: Takeover pattern: new device + new district + 02:47 + 88% balance drain to W-90022 (age 11 h). Recommended: hold, step-up verification via call-back, lock session, flag W-90022 to graph engine. Customer education message queued.

SC-03 - Prize / lottery fee scam (S2) - small amounts, dangerous counterparty

Behaviour: the amount is normal for the sender, so sender-side anomaly is low. The warning must come from the **recipient's behaviour** - the model must not rely on amount alone.

Fact	Value
Victim	Karim (C006118), 41, farmer, feature phone, median send 600 Tk.
Event	Sends 500 Tk 'verification fee' to W-41277 after an SMS 'You won a motorcycle'. W-41277 is 36 h old; 26 victims in 40 h sent 300-2,000 Tk ; collector has cashed out 87% of inflow.

Model signals: Sender anomaly low; recipient_unique_senders_48h 26; first-time ratio 0.92; recipient_age 36 h; cash-out ratio 0.87.

Fusion: $0.5 \times 0.75 + 0.3 \times 0.25 + 0.2 \times 1.00 = 0.65 \rightarrow$ **65/100, band WARN.** Counterparty graph evidence pushes it into WARN.

▼ Risk score 65/100



Customer-facing alert

Possible prize scam. Risk 65/100.

Many people have sent small amounts to this number in the last 2 days and it cashes out quickly. Real prizes never ask you to pay money first. Do you want to continue?

[Cancel] [Continue]

Bangla (transliterated): Ei number-e gato 2 dine onek manush chhoto chhoto taka pathiyeche. Asol puroshkar-e age taka chaya na. Ki korben?

Analyst case report: W-41277 identified as S2 collector. 26 victims so far. Recommended: freeze wallet, send warning SMS to all 26 senders, add number to watchlist.

SC-04 - Fake online seller (S4) - advance payment

Behaviour: looks like a small business but behaves like a collector: no supplier payments, only buyer deposits and cash-outs, no repeat buyers.

Fact	Value
Buyer	Nusrat (C008830), 24, student, smartphone, median send 900 Tk. Sees 'discount saree' page; pays 3,500 Tk advance.
Seller wallet	W-55102, registered 5 days ago. 18 first-time buyers in 4 days (500-6,000 Tk). Zero outflows to suppliers/merchants. Two cash-outs at the same agent each evening.

Model signals: recipient_age 5 d; in-degree 18; supplier_outflow_count 0; first-time ratio 1.0; repeat-buyer share 0; cash-out ratio 0.9.

Fusion: $0.5 \times 0.70 + 0.3 \times 0.30 + 0.2 \times 0.80 = 0.60 \rightarrow$ **60/100, band WARN.** Advice: use cash-on-delivery or upay Merchant Pay.

▼ Risk score 60/100



Customer-facing alert

Careful with advance payments. Risk 60/100.

This number has only 5 days of history, received money from 18 different buyers and never paid suppliers. Safer: cash-on-delivery or a verified Merchant Pay shop.

[Cancel] [Continue anyway]

Bangla (transliterated): Ei number matro 5 din dhore ache, 18 jon kreta taka diyeche kintu kono supplier-ke taka daye ni. Cash on delivery beshi nirapod.

Analyst case report: W-55102 matches fake-seller profile. 18 buyers, 0 repeat. Recommended: review merchant claim, request seller KYC, hold pending cash-outs.

SC-05 - Mule chain in 12 minutes (S3) - the graph showcase

Behaviour: each wallet alone looks like a normal pass-through, but the whole path forwards almost the full amount within minutes and ends in an agent cash-out.

Hop	Wallet (age)	Time	Amount	Note
0	Victim C001275 -> W-A (30 h)	14:02	19,500	Stolen via S1
1	W-A -> W-B (44 h)	14:06 (+4 min)	19,000	97% forwarded
2	W-B -> W-C (28 h)	14:13 (+7)	18,500	Branches 60/40
3	W-C -> Agent A0391 cash-out	14:21 (+8)	29,000 cap split in 2	Cash-out

Model signals: inflow_outflow_lag 4-8 min; forward ratio 0.97; node ages 28-44 h; path length 3 ending in cash-out; shared agent across 6 chains.

Fusion: $0.5 \times 0.93 + 0.3 \times 0.70 + 0.2 \times 1.00 = 0.88 \rightarrow$ **88/100, band HIGH / HOLD.** Alert is shown to the mule holder (hold) and to the analyst.

▼ Risk score 88/100



Customer-facing alert

Transfer on hold for security. Risk 88/100.

This transfer looks like part of a fraud chain. Your wallet is temporarily limited. Please contact upay support with your ID.

[Contact support]

Bangla (transliterated): Ei transfer-ti nirapottar karone aatke rakha hoyeche. Onugroho kore upay support-e jogajog korun.

Analyst case report: Chain: C001275 -> W-A -> W-B -> W-C -> A0391. Total 19,500 Tk in 19 min. 6 chains share A0391. Recommended: freeze W-A/W-B/W-C, review A0391, trace upstream victims.

SC-06 - Cash-out alert at the agent (your 'cash-out' question)

Behaviour: the scam money leaves the system at the cash-out. The interruption goes to the **agent screen / analyst**, not just to the wallet holder: 'Are you sure you want to cash out for this account?'

Fact	Value
Holder (fraud wallet)	W-77310 (from SC-01), age 68 h, KYC-1 on paper (documents mismatch flagged by rules).
Event	Tries cash-out 29,000 Tk at agent A0457, 6 min after 4 first-time inbound transfers totalling 31,000 Tk. Third cash-out in 2 hours.

Model signals: cashout_ratio 0.94; time since inbound 6 min; account age 68 h; KYC mismatch (amount over KYC-1 daily cap); agent has flagged-wallet share 0.31.

Fusion: $0.5 \times 0.92 + 0.3 \times 0.78 + 0.2 \times 1.00 = 0.89 \rightarrow$ **89/100, band HIGH / HOLD**. Cash-out is held until agent verifies ID or support approves.



Customer-facing alert

AGENT SCREEN - Cash-out risk HIGH (88/100).

New account (3 days), 4 first-time senders in 10 minutes, amount over KYC limit. **Verify ID and ask the customer how they know the senders before paying out.** Hold recommended.

[Hold and call upay] [Verified - release]

Bangla (transliterated): Notun account (3 din), 10 minute-e 4 jon notun preronkari, KYC limit-er beshi. Taka dewar age ID mile nin ebong upay-ke jan.

Analyst case report: Cash-out of 29,000 Tk requested at A0457 by high-risk wallet W-77310. Held. Agent A0457 flagged-wallet share 31% (peer avg 2%).

SC-07 - Rogue agent (S5): entity-level alert

Behaviour: one agent processes many more cash-outs than similar agents nearby, often for flagged wallets and just under the per-transaction cap. The alert goes to the risk analyst - the customer never sees it.

Metric (7-day)	Agent A0391	Peer group PG-DHK-U-03	Ratio / z-score
Daily cash-outs	412	46 (std 11)	9.0x / z = 33
Share from wallets < 7 days old	58%	7%	8.3x
Round amounts 29,000-30,000	44%	3%	14.7x
Wallets flagged by Prohori	31%	2%	15.5x
Invalid third-party transfers at agent	6	0	new

Agent risk score: $0.5 \times \text{z-score component} + 0.3 \times \text{flagged share} + 0.2 \times \text{invalid-activity flag} = 91/100 \rightarrow$ HIGH. **Analyst report:** A0391 cash-out volume is 9x peers with 31% flagged wallets and cap-hugging amounts. Recommended: field audit, temporarily limit cash-out cap, review CCTV/KYC records. (Use as an optional module, 'Agent Watch'.)

SC-08 - SIM-swap takeover (S6) - intentionally borderline

Behaviour: no failed PIN attempts and plausible amount, but device, hour and drain are off. Show the judges that Prohori **admits uncertainty** and chooses a gentle step-up instead of a hard block.

Fact	Value
Victim	Imran (C003340), 38, shop owner, smartphone, normal use for 14 months.
Event	Silence 6 h (phone stops working - SIM replaced by fraudster), then 05:10 new device_id, 12,000 Tk (78% of balance) to W-60877 (age 20 h).

Model signals: is_new_device=1; hour_deviation 3.9 sd; drain 0.78; recipient age 20 h; no pin errors.

Fusion: $0.5 \times 0.60 + 0.3 \times 0.75 + 0.2 \times 0.30 = 0.59 \rightarrow$ **59/100, band NUDGE**. Mid score = step-up verification, not hard block (responsible-AI point).



Customer-facing alert

Verify it is you. Risk 59/100.

We noticed a new phone and an unusual time. Please answer your security question to continue.

[Verify] [Cancel]

Bangla (transliterated): Notun phone ebong osadharon समय. Cholte hole nirapotta proshno-r uttor din.

Analyst case report: Medium-risk takeover candidate. Recommended: step-up verification and 24 h monitoring; if not verified within 15 min, escalate to WARN.

SC-09 - Guided victim: coached by a caller (S7)

Behaviour: long confused session, repeated PIN errors, large amount for this user, young recipient. The best protection is a very plain, short voice/text message because the user may not use the app.

Fact	Value
Victim	Rokeya (C005901), 67, housewife, USSD feature phone, median send 400 Tk, KYC-1.
Event	Tries 6,000 Tk (15x median) to W-83019 (age 9 h). Session 412 s (normal 35 s). pin_attempts = 3 . She had a call during the session; afterwards tries cash-out at agent.

Model signals: session_seconds 412; pin_attempts 3; amount_ratio 15; recipient_age 9 h; first-time counterparty; USSD channel.

Fusion: $0.5 \times 0.72 + 0.3 \times 0.70 + 0.2 \times 0.40 = 0.65 \rightarrow$ **65/100, band WARN**. Voice message plus 'hang up the call' advice.

▼ Risk score 65/100



Customer-facing alert

STOP. Is someone on the phone telling you what to do? Risk 65/100. upay never asks you to send money by phone. Hang up and ask a family member.

[Cancel] [Continue]

Bangla (transliterated): Thamun! Keu ki phone-e bole dichhe ki korte hobe? upay kokhono phone-e taka pathate bole na. Phone kete din.

Analyst case report: Coached-victim signature (S7). Recommended: delay 10 min, SMS to registered family contact if available, tag customer for education call.

SC-10 - Structured amounts / sibling wallets (S3 variant)

Behaviour: amounts are kept just under limits and spread over sibling wallets to avoid single-transaction rules; graph clustering by shared agent and device reveals the ring.

Fact	Value
Wallet	W-31905, registered 2 days ago as a normal wallet (fraudster-controlled, 3 coordinated siblings W-31905/06/07).
Event	Smurfing: 11 inbound of 4,900-9,900 Tk from 11 senders (all just under KYC-1 per-day reception amounts), then 3 outward transfers to 3 sibling wallets of 9,500 Tk each within 20 min.

Model signals: near-cap amounts (0.95-0.99 of cap); fan-in 11 then fan-out 3 within 20 min; shared device_id across 3 wallets; node ages under 72 h.

Fusion: $0.5 \times 0.80 + 0.3 \times 0.60 + 0.2 \times 1.00 = 0.78 \rightarrow$ **78/100, band WARN**. Hold sibling transfers, escalate cluster.

▼ Risk score 78/100



Customer-facing alert

Transfer on hold for review. Risk 78/100. Your wallet shows an unusual pattern. Please verify your identity.

[Contact support]

Bangla (transliterated): Apnar wallet-e osadharon lenden dhora poreche. Porichoy jachai korun.

Analyst case report: Ring of 3 wallets sharing device D-9917 and agent A0391. Recommended: freeze ring, review KYC.

Part 7 - Benign look-alikes (must NOT alarm)

A fraud system that cries wolf is useless. Plant these legitimate cases in every split, label them `is_benign_lookalike=1` and include them in acceptance tests. Scores are expected values from the same fusion formula.

ID	Case	What happens	P	Anom	Graph	Score	Band
SB-01	Eid gift to mother	Customer sends 15,000 Tk to a contact used 40 times in 2 years during Eid week	0.05	0.20	0.0	9	ALLOW
SB-02	Salary day	Garment worker cashes in 18,000 Tk on the 3rd and cashes out 12,000 Tk in two steps	0.04	0.10	0.0	5	ALLOW
SB-03	Tuition fee	Student pays 22,000 Tk to a verified university merchant wallet (new to sender, merchant registered 3 years)	0.10	0.35	0.0	16	ALLOW
SB-04	Market-day shop	Shop wallet (age 14 months, merchant tag) receives 45 payments from 38 senders and cashes out in the evening	0.08	0.30	0.1	15	ALLOW
SB-05	Spouse's new wallet	Husband sends 12,000 Tk to wife's wallet registered yesterday, with phone-contact link and shared home agent	0.30	0.35	0.0	26	ALLOW
SB-06	Rare large send	Regular customer sends 25,000 Tk to a known contact for medical emergency (daytime, usual device)	0.18	0.60	0.0	27	ALLOW
SB-07	Remittance day	Remittance from abroad (REMITTANCE_IN) to a rural wallet, then immediate cash-out of 90%	0.06	0.30	0.0	12	ALLOW
SB-08	Government allowance	G2P disbursement to 300 wallets from one trusted source id	0.02	0.10	0.0	4	ALLOW

Design note: SB-05 deliberately lands in NUDGE (a friendly 'is this your family member?' prompt). A small friction on a legitimate case is acceptable and shows honest trade-offs; target a false-alarm rate (WARN or HIGH on legit rows) below 1-2% and report it.

Part 8 - Risk policy, warning UX and acceptance tests

8.1 Bands and what the user sees

Score	Band	Customer experience	System action
0-29	ALLOW	Normal flow, no interruption	Log only
30-59	NUDGE	One-line friendly hint ('first time sending to this number - double-check') or step-up verification for new device	Log; monitor
60-84	WARN	Full warning with reasons + 'Do you know this person?' + Cancel is the default button	Alert to analyst queue
85-100	HIGH / HOLD	Strong warning, extra confirmation and 10-minute cooling-off or call-back; cash-out held at agent	High-priority alert; freeze recommendation (human approves)

Fusion weights (0.5 / 0.3 / 0.2) and thresholds (30 / 60 / 85) are starting points - tune on the validation set, keep them in `config.json`, and state the chosen operating point (recall at a fixed false-alarm rate) in your pitch. Humans make the final decision on blocks.

8.2 Warning message template (works for app, USSD and voice)

- **Header:** short and calm - 'Stop and check' / 'Careful'.
- **Risk:** 'Risk 89/100 (HIGH)'. Use 'Fraud probability' only after calibration.
- **Reasons (max 3):** plain facts - wallet age, number of strangers who paid it, amount vs your usual.
- **Question:** 'Do you personally know and trust this person?'
- **Authority line:** 'upay never asks for PIN or OTP by phone.'
- **Buttons:** Cancel (default, large) / Continue (smaller, needs extra step on HIGH).
- **USSD / feature phone:** max 160 characters + voice playback; **Bangla** first, English second.

8.3 Acceptance tests - run these on the generated files BEFORE training

#	Test	Pass condition
T1	Row count and window	550k-650k rows; timestamps within 60 days; sorted
T2	Fraud prevalence	0.4%-0.6% of rows; class mix within +/-5 points of Part 2.1
T3	Balance integrity	For every row: after = before - amount - fee; no negative balance (except labelled violations)
T4	KYC caps	99% of normal rows within caps; violations only in labelled fraud/benign cases
T5	Normal amount shape	Median P2P amount 900-2,500 Tk; under 1% of normal sends above 15,000 Tk
T6	Calendar effects	Friday uplift 20-50%; days 1-5 cash-in uplift 1.5-2.2x; Eid window volume 2-3x
T7	Night activity	Normal rows 00-05 under 2%; S1 rows 00-05 over 50%
T8	Scenario presence	All SC-01..SC-10 and SB-01..SB-08 exist with expected_band set and the stated facts reproducible
T9	Collector structure	Each S2/S3/S4 collector has recipient_age under 72h at first inbound and over 10 unique senders; at least 80% first-time
T10	Agent outliers	Rogue agents are 7-10x peers; no normal agent above 2x
T11	No leakage	Raw files contain no label-derived columns; derived features use only past data (verify with a shift test)
T12	Benign look-alikes	At least 1.5x as many benign look-alike cases as fraud cases; baseline model scores them below 60 for 98%
T13	Reproducibility	Same seed gives byte-identical CSVs
T14	Privacy	No real phone/NID patterns; all names/ids synthetic; README disclosure written

Part 9 - Generator prompt for Claude and verification checklist

Paste the prompt below into Claude (together with Parts 3-5 and the scenario tables). Ask for the script in modules so you can read and understand every part - the rulebook expects you to explain your code.

Write a Python 3 script (pandas, numpy, faker optional, seed=42) that generates a synthetic Bangladesh mobile-financial-service dataset for a fraud-detection hackathon.

OUTPUT FILES: customers.csv, agents.csv, transactions.csv, labels.csv, config.yaml.
Use the schemas in the attached spec exactly. All data must be fake.

STRUCTURE (separate modules/functions):

1. config_loader (KYC caps, fees, volumes, class mix are config, not hard-coded)
2. build_customers(), build_agents(), build_profiles()
3. simulate_normal_life() # follows Part 4: lognormal amounts by income band, # regular contacts, Friday/salary/Eid/bill calendar, # new legit wallets, merchants, remittance and G2P
4. inject_s1()...inject_s7() # follow Part 5 recipes with randomised ranges
5. plant_scenarios() # SC-01..SC-10 and SB-01..SB-08 with exact facts
6. enforce_balances() # ledger consistency, KYC caps, fee rules
7. write_files() # labels.csv kept separate; no leakage columns
8. run_tests() # implement T1-T14 and print PASS/FAIL

RULES: time-sorted rows; balances consistent; fraud rows labelled with class, role, case_id; ~20% of fraud cases weak/ambiguous; benign look-alikes 1.5x fraud cases; amounts are Tk (BDT); timezone Asia/Dhaka; print a summary report (row counts, fraud rate, class mix, amount histogram, scenario presence).

9.1 After generation - checklist

- Run T1-T14 and keep the output in your repo as docs/data_validation.txt.
- Plot: amount histogram (log scale), transactions by hour, by weekday, by day-of-month. Compare with Part 4 expectations.
- Open SC-01 in the data: confirm you can reproduce the 'median 2,000 Tk, tries 20,000 Tk, recipient 68 h old, 29 unique senders' story from raw rows alone.
- Train a quick baseline (logistic regression on 6 features) to confirm the data is learnable but not trivial: PR-AUC should be clearly above 0.5 and below 0.99.
- Write a Dataset Disclosure paragraph in the README: synthetic, generator logic, over-represented fraud, limitations, 'as of' date for upay/Bangladesh Bank rules.

9.2 What to verify and where

Topic	Where to check
upay limits and charges (P2P, cash-out, add money)	upay's official limits-and-charges page and in-app terms; screenshot with date
Bangladesh Bank MFS Regulations (agents, KYC, monitoring)	Bangladesh Bank website: MFS Regulations 2022 and later circulars
Fraud statistics (PRI survey, 1 in 10 victims, 56% impersonation)	PRI Bangladesh publication on MFS fraud; quote with page number
Fraud typologies and BFIU reporting	BFIU guidelines and Bangladesh Bank AML/CFT circulars
PaySim (optional benchmark data)	Kaggle: ealaxi/paysim1 - use only as a technical benchmark, not as Bangladesh behaviour

Bangla UI strings are supplied separately as prohori_warnings_bn.json (UTF-8), because this PDF's font cannot shape Bangla script correctly; the transliterations above are only for reading.